Dual-ligand screen / Bench sheet

One run · 48 pairs · 30 ligands · 100 µL reactions

95 °C / 500 rpm / 4 h. 0.1 M SM; 6 mol% each ligand; 12 mol% Pd; naphthalene 10 mol%. One program runs from stocks through sampling. Source vials are glass, in the supplied HTEL holder.

Slot	Load
1	HTEL stock holder: map on page 2. Moves 1 → 10 → 1 before dosing.
2	One prefilled 12-channel reservoir; loads below.
3	Para-Dox block with 48 empty reaction vials.
5, 8	300 µL tips for LEFT p300 single GEN2.
6	One empty 96-well LC plate; retain it throughout.
9	300 µL tips for RIGHT p300 eight-channel GEN2.
10	Heater-shaker / universal flat adapter. Slots 4, 7 and 11 empty.

Bulk liquids / reservoir

Quench, 40 mL: 36 mL ACN + 3.95 mL water + 50 µL formic acid. **LC diluent, 60 mL:** 30 mL ACN + 29.94 mL water + 60 µL formic acid.

Channels	Liquid	Load per channel
A1-A2	Toluene	9 mL
A3-A5	Quench	9 mL
A6-A10	LC diluent	10 mL
A11 / A12	ACN / empty	3.5 mL / empty

Run in this order

1. Weigh each ligand once. Load all vials/liquids. Import **run_screen.py**; check offsets and simulate. Robot prepares stocks. Keep ligand vials A1-F5 plus A6/B6 **OPEN**, analytical vials capped. Move holder 1 → 10; shake **750 rpm / 2 min, heater OFF**; return to 1. Robot doses 20 µL SM/IS + 30 µL A + 30 µL B + 20 µL Pd.

2. Hand-cap reaction vials. Used reaction stocks stay open; analytical vials stay capped until needed. Cover reservoir/LC. Move block 3 → 10. Start timer: **4 h is nominal**. At 3 h, uncover reservoir/LC and uncap analytical vials. Robot prepares standards A7-G7; cover reservoir/LC again. Used analytical vials stay open. For a longer hold, wait at this post-standards pause, then resume. Longer holds/pre-quench waits are allowed and logged.

3. Heater/shaker OFF. Move **sealed** block 10 → 3; cool passively. **Confirm block/vials ≤30 °C with a contact probe before opening.** App also waits for module ≤30 °C. Resume same program. Robot preloads 480 µL LC diluent/well. Uncap when prompted; add 400 µL quench/vial. Open block: 750 rpm / 2 min, heater OFF; return to 3. Fresh-tip sample: 20 µL/well.

4. Recap reaction vials. LC plate: 6 → cooled 10; shake uncovered 750 rpm / 2 min, heater OFF. Seal and hand to LC once: seven standards FIRST, then 48 samples. Preserve actual injection map, original LC/MS files and robot logs.

Margins: source vials ≤1.6 mL; ≥150 µL left. Reservoir channels retain ≥1.5 mL. Reaction vials end at 500 µL; LC wells at 500 µL. Aspiration 1.5 mm above bottom. Pause for liquid-handling faults.

Weighing / stock-holder positions

All positions are in the HTEL stock holder in **slot 1**. Ligands occupy columns 1-5. Column 8 stays empty. The robot adds solvent directly in these vials. No stock transfers or refills.

Vial	Label	Material	mg
A1	L01	MorDal Phos	2.4
B1	L02	racBINAP	3.2
C1	L03	RockPhos	2.1
D1	L04	Alphos	4.2
E1	L05	AdBrettPhos	4.6
F1	L06	Me4t-BuXPhos	2.2
A2	L08	t-BuDave Phos	2.0
B2	L09	Xantphos	2.6
C2	L10	OTIPS DalPhos	2.1
D2	L11	Me3OMe-t-BuXPhos	2.0
E2	L12	VBRIDP	2.0
F2	L13	Trimesityl phosphine	2.0
A3	L14	CataCXium ABn	2.0
B3	L15	dppb	2.8
C3	L16	dcypf	3.8
D3	L17	t-BuBrettPhos	3.2
E3	L18	dtbpf	3.8
F3	L19	dppp	2.4
A4	L20	dppf	3.3
B4	L21	Xphos	2.0
C4	L22	RuPhos	2.1
D4	L23	SPhos	2.0
E4	L24	BrettPhos	2.4
F4	L25	MeDalPhos	2.5
A5	L26	Taniaphos SL-T001-2	3.5
B5	L27	tris(3-chlorophenyl)phosphine	2.7
C5	L28	BippyPhos	4.4
D5	L29	meCgPPh	2.0
E5	L30	N-XantPhos	2.5
F5	L31	Trixie Phos	2.0
A6	SM_IS	Pyridine-2-sulfonyl fluoride + naphthalene	96.7 SM +7.7 IS
B6	Pd	Pd(COD)(DQ)	27.3
C6	IS_MASTER	Naphthalene (analytical master)	10.0
D6	SM_FIT	Pyridine-2-sulfonyl fluoride	5.0
E6	SM_CHECK	Pyridine-2-sulfonyl fluoride	5.0
F6	P_FIT	2-Fluoropyridine	5.0
A7	P_CHECK	2-Fluoropyridine	5.0

Five empty working vials: B7 HIGH, C7 LOW, D7 CHECK, E7 IS_WORK, F7 MATRIX. Weigh FIT/CHECK standards separately; weigh liquid product by difference. Weigh ligands in the glovebox and retain original reagent bottles there.

Full procedure, maps, Python files and LC instructions: github.com/MiquelAngelPerezPuigdo/Agentic-AI-for-Dual-Ligand-Discovery / run